

RBC Eire I

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In what follows I set out a two-sector, small open economy RBC model – supplemented by fiscal policy and long run growth – of the Irish economy. I simulate the model in R using the gEcon package and fit it to Irish macroeconomic time-series data. The model (when stationarized) is BGP consistent, satisfies BK conditions and is solvable.

1 Model Set-up

1.1 Assumptions

1. Markets are perfectly competitive.
2. Prices and wages are fully flexible.
3. The model is in discrete time with an infinite number of periods.
4. Agents hold rational expectations and adhere to a choice theoretic framework.
5. Long run exogenous economic growth.
6. Balanced growth path.
 - 1-6 are textbook RBC assumptions.
7. The consumer side of the economy is modeled by a representative household with Jaimovich-Rebelo (JR) preferences.
 - JR preferences can be parameterized to nullify wealth effects on labor supply and are BGP consistent.
8. The production side of the economy is modeled by two sectors (domestic and multinational).
 - Multinationals play an out-sized role in the Irish economy and exhibit highly idiosyncratic behavior. For this reason, I model it as a separate sector to other economic activity.
9. Each sector exhibits CRS Cobb-Douglas production technology with Harrod-Neutral technical progress.
 - CRS Cobb-Douglas is chosen for the standard reasons. Harrod-neutral technical progress is chosen to facilitate stationarizing the model.
10. Multinational investment is exogenously determined.
11. Labor is imperfectly mobile across the domestic and multinational sectors.
12. Capital is perfectly *immobile* across each sector.
13. Government expenditures are funded by ad valorem taxes on domestic and multinational labor income and multinational capital income.
 - The vast majority of Irish government distortionary tax revenues are raised from a combination of taxes on labor income and multinational capital income. I therefore omit capital income taxes on non-multinational firms as they make up a negligible portion of the Irish tax base.
14. Labor income tax is determined endogenously; multinational capital income tax is determined exogenously.

15. Government borrows from international investors.
16. Government follows Barro-Bohn fiscal policy rule.
17. Small open economy: 100% of Multinational sector output exported; 100% of Domestic sector output consumed domestically; 100% of net Multinational sector capital income repatriated abroad; Representative Household consumes imported good; 100% of Multinational investment funded by foreign investors; Representative Household lends/borrows on international financial markets; 100% of Domestic investment funded by Representative household; and 100% of domestic net labor and capital income output accumulated by Representative Household
18. The economy is subject to stochastic exogenous shocks to: (i) domestic sector productivity; (ii) multinational sector productivity; (iii) government spending; (iv) imported good prices; (v) exported good prices; (vi) multinational investment; and (vii) the world interest rate $r_{\text{world}t}$.

2 Model in Levels

2.1 The Representative Household

CES Consumption Aggregator:

$$C_t = \left[\theta \cdot C_{Dt}^{\frac{\eta-1}{\eta}} + (1-\theta) \cdot C_{Mt}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (1)$$

JR Preferences:

$$U_t(C_t, L_t, S_t) = \frac{(C_t - \psi L_t^\mu S_t)^{1-\sigma} - 1}{1-\sigma}, \quad S_t = C_t^\xi ((1+\gamma)S_{t-1})^{1-\xi} \quad (2)$$

Lifetime Utility:

$$U = \mathbb{E}_0 \left[\sum_{t=0}^{+\infty} \beta^t U_t \right] \quad (3)$$

Household Budget Constraint (Domestic Good is Numeraire):

$$C_{Dt} + p_{Mt}C_{Mt} + I_{Dt} + B_t = (1 + r_{Bt-1})B_{t-1} + (1 - \tau_{Lt})W_t L_t + r_{Dt}K_{Dt} \quad (4)$$

2.2 Two Sectors

$$Y_{GDPt} = Y_{Dt} + p_{Xt}Y_{Mt} \quad (5)$$

Domestic Production (D) with CRS Harrod-neutral Cobb-Douglas Technology:

$$Y_{Dt} = K_{Dt}^\alpha (A_{Dt}L_{Dt})^{1-\alpha} \quad (6)$$

$$r_{Dt} = \alpha \cdot \frac{Y_{Dt}}{K_{Dt}} \quad (7)$$

$$W_{Dt} = (1 - \alpha) \cdot \frac{Y_{Dt}}{L_{Dt}} \quad (8)$$

$$K_{Dt+1} = I_{Dt} + (1 - \delta_D)K_{Dt} \quad (9)$$

Multinational Production (M) with CRS Harrod-neutral Cobb-Douglas Technology:

$$Y_{Mt} = K_{Mt}^\nu (A_{Mt} L_{Mt})^{1-\nu} \quad (10)$$

$$r_{Mt} = \nu \cdot \frac{p_{Xt} Y_{Mt}}{K_{Mt}} \quad (11)$$

$$W_{Mt} = (1 - \nu) \cdot \frac{p_{Xt} Y_{Mt}}{L_{Mt}} \quad (12)$$

$$K_{Mt+1} = I_{Mt} + (1 - \delta_M) K_{Mt} \quad (13)$$

Imperfectly Mobile Labor Supply:

$$L_t = L_{Dt} + L_{Mt} \quad (14)$$

$$\Phi_t = \frac{\phi_L}{2} (L_{Mt} - L_{Mt-1})^2 \quad (15)$$

2.3 Government

Budgetary Constraint:

$$G_t + (1 + r_{\text{world}t-1}) D_{t-1} = \tau_{Lt} W_t L_t + \tau_{Mt} r_{Mt} K_{Mt} + D_t \quad (16)$$

Barro-Bohn Fiscal Rule with Endogenous Labor Income Tax:

$$\tau_{Lt} = \bar{\tau}_L + \phi_D \cdot \left(\frac{D_{t-1}}{GNI_{t-1}^*} - \bar{d}^* \right) + \phi_y \cdot \hat{Y}_{t-1}^* \quad (17)$$

$$\hat{Y}_t^* = \ln Y_{GNI^*t} - \ln \bar{Y}_{GNI^*t}, \quad \bar{Y}_{GNI^*t} = (1 + \gamma) \bar{Y}_{GNI^*t-1} \quad (18)$$

Multinational Capital Income tax is Exogenous:

$$\tau_{Mt} = \bar{\tau}_M \quad (19)$$

2.4 Small Open Economy

Exports (100% of Multinational Output is Exported):

$$Y_{Mt} = X_t \quad (20)$$

Open Economy:

$$(B_t - D_t) - (B_{t-1} - D_{t-1}) = C A_t + p_{mt} I_{Mt} \quad (21)$$

$$C A_t = N X_t + r_{Bt-1} B_{t-1} - r_{\text{world}t-1} D_{t-1} - (1 - \tau_{Mt}) \cdot r_{Mt} K_{Mt} \quad (22)$$

$$N X_t = p_{Xt} X_t - p_{mt} C_{mt} - p_{mt} I_{Mt} \quad (23)$$

$$B < 0 \rightarrow \text{Representative Household is a debtor} \quad (24)$$

$$B > 0 \rightarrow \text{Representative Household is a creditor} \quad (25)$$

Debt-Elastic Interest Rate on Foreign Assets:

$$r_{Bt} = r_{\text{world}t} + \phi_B \cdot (e^{\bar{b}-b_t} - 1), \quad b_t = \frac{B_t}{\Gamma_t} \quad (26)$$

2.5 Aggregate Output

$$Y_{GDPt} = C_{Dt} + p_{mt} C_{mt} + G_t + I_{Dt} + p_{mt} I_{Mt} + N X_t \quad (27)$$

$$Y_{GNI^*t} = Y_{GDPt} + r_{Bt-1} B_{t-1} - r_{\text{world}t-1} D_{t-1} - (1 - \tau_{Mt}) r_{Mt} K_{Mt} \quad (28)$$

$$Y_{Dt} = C_{Dt} + I_{Dt} + G_t + \Phi_t \quad (29)$$

2.6 Stochastic Exogenous BGP Consistent Shock Processes

$$\Gamma_t = (1 + \gamma)^t \quad (30)$$

$$\ln Z_{I_M t} = \rho_{I_M} \ln Z_{I_M t-1} + \varepsilon_{I_M t}, \quad I_{M t} = \zeta_I \cdot \Gamma_t Z_{I_M t} \quad (31)$$

$$\ln Z_{D t} = \rho_D \ln Z_{D t-1} + \varepsilon_{D t}, \quad A_{D t} = \Gamma_t Z_{D t}, \quad (32)$$

$$\ln Z_{M t} = \rho_M \ln Z_{M t-1} + \varepsilon_{M t}, \quad A_{M t} = \zeta_M \cdot \Gamma_t Z_{M t} \quad (33)$$

$$\ln Z_{G t} = \rho_G \ln Z_{G t-1} + \varepsilon_{G t}, \quad G_t = \zeta_G \cdot \Gamma_t Z_{G t} \quad (34)$$

$$\ln p_{m t} = \rho_m \ln p_{m t-1} + \varepsilon_{m t} \quad (35)$$

$$\ln p_{X t} = \rho_X \ln p_{X t-1} + \varepsilon_{X t} \quad (36)$$

$$\ln(1 + r_{\text{world}t}) = (1 - \rho_{r_{\text{world}}}) \ln(1 + \bar{r}_{\text{world}}) + \rho_{r_{\text{world}}} \ln(1 + r_{\text{world}t-1}) + \varepsilon_{r_{\text{world}t}} \quad (37)$$

3 Stationarized Model

3.1 Conventions

Henceforth:

$$j_t \equiv \frac{J_t}{\Gamma_t} \quad \text{for all trended variables } J_t \quad (38)$$

$$\therefore (1 + \gamma) \cdot j_{t+1} = \frac{J_{t+1}}{\Gamma_t} \quad (39)$$

De facto stationarized variables:

$$L_t, L_{D t}, L_{M t}, r_{\text{world}t}, r_{M t}, r_{B t}, p_{m t}, p_{X t}, \tau_{L t}, \tau_{M t}, Z_{D t}, Z_{M t}, Z_{G t}, Z_{I_M t}.$$

3.2 The Representative Household

$$c_t = \left[\theta \cdot c_{D t}^{\frac{\eta-1}{\eta}} + (1 - \theta) \cdot c_{m t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (40)$$

$$u_t(c_t, L_t, s_t) = \frac{(c_t - \psi L_t^\mu s_t)^{1-\sigma} - 1}{1 - \sigma}, \quad s_t = c_t^\xi s_{t-1}^{1-\xi} \quad (41)$$

$$u = \mathbb{E}_0 \left[\sum_{t=0}^{+\infty} \tilde{\beta}^t u_t \right], \quad \tilde{\beta} = \beta(1 + \gamma)^{1-\sigma} \quad (42)$$

$$c_{D t} + p_{m t} c_{m t} + i_{D t} + b_t = \frac{(1 + r_{B t-1}) b_{t-1}}{1 + \gamma} + (1 - \tau_{L t}) w_t L_t + r_{D t} k_{D t} \quad (43)$$

$$(44)$$

3.3 Two Sectors

$$y_{\text{GDP}t} = y_{\text{Dt}} + p_{\text{X}t}y_{\text{Mt}} \quad (45)$$

$$y_{\text{Dt}} = k_{\text{Dt}}^\alpha (Z_{\text{Dt}}L_{\text{Dt}})^{1-\alpha} \quad (46)$$

$$r_{\text{Dt}} = \alpha \cdot \frac{y_{\text{Dt}}}{k_{\text{Dt}}} \quad (47)$$

$$w_t = (1 - \alpha) \cdot \frac{y_{\text{Dt}}}{L_{\text{Dt}}} \quad (48)$$

$$(1 + \gamma)k_{\text{Dt}+1} = i_{\text{Dt}} + (1 - \delta_{\text{D}})k_{\text{Dt}} \quad (49)$$

$$y_{\text{Mt}} = k_{\text{Mt}}^\nu (\zeta_{\text{M}}Z_{\text{Mt}}L_{\text{Mt}})^{1-\nu} \quad (50)$$

$$r_{\text{Mt}} = \nu \cdot \frac{p_{\text{X}t}y_{\text{Mt}}}{k_{\text{Mt}}} \quad (51)$$

$$w_t = (1 - \nu) \cdot \frac{p_{\text{X}t}y_{\text{Mt}}}{L_{\text{Mt}}} \quad (52)$$

$$(1 + \gamma)k_{\text{Mt}+1} = i_{\text{Mt}} + (1 - \delta_{\text{M}})k_{\text{Mt}} \quad (53)$$

$$L_t = L_{\text{Dt}} + L_{\text{Mt}} \quad (54)$$

$$\Phi_t = \frac{\phi_L}{2}(L_{\text{Mt}} - L_{\text{Mt}-1})^2 \quad (55)$$

3.4 Government

$$g_t + \frac{(1 + r_{\text{world}t-1})d_{t-1}}{1 + \gamma} = \tau_{\text{Lt}}w_tL_t + \tau_{\text{Mt}}r_{\text{Mt}}k_{\text{Mt}} + d_t \quad (56)$$

$$\tau_{\text{Lt}} = \bar{\tau}_{\text{L}} + \phi_{\text{D}} \cdot \left(\frac{d_{t-1}}{gn_{t-1}^*} - \bar{d}^* \right) + \phi_y \cdot \hat{y}_{t-1}^*, \quad \hat{y}_t^* = \ln y_{\text{GNI}^*t} - \ln \bar{y}_{\text{GNI}^*} \quad (57)$$

$$\tau_{\text{Mt}} = \bar{\tau}_{\text{M}} \quad (58)$$

3.5 Small Open Economy

$$y_{\text{Mt}} = x_t \quad (59)$$

$$(b_t - d_t) - \frac{(b_{t-1} - d_{t-1})}{1 + \gamma} = ca_t + p_{\text{mt}}i_{\text{Mt}} \quad (60)$$

$$ca_t = nx_t + \frac{r_{\text{B}t-1}b_{t-1}}{1 + \gamma} - \frac{r_{\text{world}t-1}d_{t-1}}{1 + \gamma} - (1 - \tau_{\text{Mt}}) \cdot r_{\text{Mt}}k_{\text{Mt}} \quad (61)$$

$$nx_t = p_{\text{X}t}x_t - p_{\text{mt}}c_{\text{mt}} - p_{\text{mt}}i_{\text{Mt}} \quad (62)$$

$$r_{\text{B}t} = r_{\text{world}t} + \phi_{\text{B}} \cdot (e^{\bar{b}-b_t} - 1) \quad (63)$$

3.6 Aggregate Output

$$y_{\text{GDP}t} = c_{\text{Dt}} + p_{\text{mt}}c_{\text{mt}} + g_t + i_{\text{Dt}} + p_{\text{mt}}i_{\text{Mt}} + nx_t \quad (64)$$

$$y_{\text{GNI}^*t} = y_{\text{GDP}t} + \frac{r_{\text{B}t-1}b_{t-1}}{1 + \gamma} - \frac{r_{\text{world}t-1}d_{t-1}}{1 + \gamma} - (1 - \tau_{\text{Mt}})r_{\text{Mt}}k_{\text{Mt}} \quad (65)$$

$$y_{\text{Dt}} = c_{\text{Dt}} + i_{\text{Dt}} + g_t + \Phi_t \quad (66)$$

3.7 Stochastic Exogenous BGP Consistent Shock Processes

$$i_{Mt} = \zeta_I Z_{I_M t} \quad (67)$$

$$g_t = \zeta_G Z_{Gt} \quad (68)$$

$$\ln Z_{I_M t} = \rho_{I_M} \ln Z_{I_M t-1} + \varepsilon_{I_M t} \quad (69)$$

$$\ln Z_{Dt} = \rho_D \ln Z_{Dt-1} + \varepsilon_{Dt} \quad (70)$$

$$\ln Z_{Mt} = \rho_M \ln Z_{Mt-1} + \varepsilon_{Mt} \quad (71)$$

$$\ln Z_{Gt} = \rho_G \ln Z_{Gt-1} + \varepsilon_{Gt} \quad (72)$$

$$\ln p_{mt} = \rho_m \ln p_{mt-1} + \varepsilon_{mt} \quad (73)$$

$$\ln p_{Xt} = \rho_X \ln p_{Xt-1} + \varepsilon_{Xt} \quad (74)$$

$$\ln(1 + r_{\text{world}t}) = (1 - \rho_{r_{\text{world}}}) \ln(1 + \bar{r}_{\text{world}}) + \rho_{r_{\text{world}}} \ln(1 + r_{\text{world}t-1}) + \varepsilon_{r_{\text{world}t}} \quad (75)$$

4 gEcon Model Specification

4.1 block REPRESENTATIVE_HOUSEHOLD

Definitions:

$$u_t = \frac{(c_t - \psi L_{st}^\mu s_t)^{1-\sigma} - 1}{1-\sigma} \quad (76)$$

$$\tilde{\beta} = \beta(1 + \gamma)^{1-\sigma} \quad (77)$$

Controls:

$$\{c_t, c_{Dt}, c_{mt}, s_t, L_{st}, k_{Dst}, i_{Dt}, b_t\}$$

Objective:

$$U_t = u_t + \tilde{\beta} \mathbb{E}_t[U_{t+1}] \quad (78)$$

Constraints:

$$c_t = \left(\theta c_{Dt}^{\frac{\eta-1}{\eta}} + (1-\theta) c_{mt}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \quad (79)$$

$$s_t = c_t^\xi s_{t-1}^{1-\xi} \quad (80)$$

$$c_{Dt} + p_{mt} c_{mt} + i_{Dt} + b_t = \frac{(1 + r_{Bt-1}) b_{t-1}}{1 + \gamma} + (1 - \tau_{Lt}) w_t L_{st} + r_{Dt} k_{Dst-1} \quad (81)$$

$$(1 + \gamma) k_{Dst} = i_{Dt} + (1 - \delta_D) k_{Dst-1} \quad (82)$$

4.2 block DOMESTIC_SECTOR

Controls:

$$\{k_{Ddt}, L_{Ddt}, y_{Dt}, \pi_{Dt}\}$$

Objective:

$$\Pi_{Dt} = \pi_{Dt} \quad (83)$$

Constraints:

$$y_{Dt} = k_{Ddt}^\alpha (Z_{Dt} L_{Ddt})^{1-\alpha} \quad (84)$$

$$\pi_{Dt} = y_{Dt} - w_t L_{Ddt} - r_{Dt} k_{Ddt} \quad (85)$$

4.3 block MULTINATIONAL_SECTOR

Identities:

$$y_{Mt} = k_{Mt-1}^\nu (\zeta_M Z_{Mt} L_{Mt})^{1-\nu} \quad (86)$$

$$r_{Mt} = \nu \frac{p_{Xt} y_{Mt}}{k_{Mt-1}} \quad (87)$$

$$(1 + \gamma) k_{Mt} = i_{Mt} + (1 - \delta_M) k_{Mt-1} \quad (88)$$

4.4 block GOVERNMENT

Identities:

$$g_t + \frac{(1 + r_{\text{world}t-1}) d_{t-1}}{1 + \gamma} = \tau_{Lt} w_t L_{st} + \bar{\tau}_M r_{Mt} k_{Mt-1} + d_t \quad (89)$$

$$\tau_{Lt} = \bar{\tau}_L + \phi_D \left(\frac{d_{t-1}}{y_{GNI t-1}^*} - \bar{d}^* \right) + \phi_y \hat{y}_{t-1}^* \quad (90)$$

$$\hat{y}_t^* = \log(y_{GNI t}^*) - \log(\bar{y}_{GNI}^*) \quad (91)$$

4.5 block EQUILIBRIUM

Identities:

$$y_{GDPt} = y_{Dt} + p_{Xt} y_{Mt} \quad (92)$$

$$y_{GNI t}^* = y_{GDPt} + \frac{r_{Bt-1} b_{t-1}}{1 + \gamma} - \frac{r_{\text{world}t-1} d_{t-1}}{1 + \gamma} - (1 - \bar{\tau}_M) r_{Mt} k_{Mt-1} \quad (93)$$

$$\Phi_t = \frac{\phi_L}{2} (L_{Mt} - L_{Mt-1})^2 \quad (94)$$

$$y_{Dt} = c_{Dt} + i_{Dt} + g_t + \Phi_t \quad (95)$$

$$k_{Ddt} = k_{Dst-1} \quad (96)$$

$$L_{st} = L_{Ddt} + L_{Mt} \quad (97)$$

$$r_{Bt} = r_{\text{world}t} + \phi_B (\exp(\bar{b} - b_t) - 1) \quad (98)$$

4.6 block EXOGENOUS_SHOCKS

Identities:

$$i_{Mt} = \zeta_{IM} Z_{IMt} \quad (99)$$

$$g_t = \zeta_G Z_{Gt} \quad (100)$$

$$\log(Z_{IMt}) = \rho_{IM} \log(Z_{IMt-1}) + \varepsilon_{IMt} \quad (101)$$

$$\log(Z_{Dt}) = \rho_D \log(Z_{Dt-1}) + \varepsilon_{Dt} \quad (102)$$

$$\log(Z_{Mt}) = \rho_M \log(Z_{Mt-1}) + \varepsilon_{Mt} \quad (103)$$

$$\log(Z_{Gt}) = \rho_G \log(Z_{Gt-1}) + \varepsilon_{Gt} \quad (104)$$

$$\log(p_{mt}) = \rho_m \log(p_{mt-1}) + \varepsilon_{mt} \quad (105)$$

$$\log(p_{Xt}) = \rho_X \log(p_{Xt-1}) + \varepsilon_{Xt} \quad (106)$$

$$\log(1 + r_{\text{world}t}) = (1 - \rho_{r_{\text{world}}}) \log(1 + \bar{r}_{\text{world}}) + \rho_{r_{\text{world}}} \log(1 + r_{\text{world}t-1}) + \varepsilon_{r_{\text{world}t}} \quad (107)$$

Shocks:

$$\{\varepsilon_{IMt}, \varepsilon_{Dt}, \varepsilon_{Mt}, \varepsilon_{Gt}, \varepsilon_{mt}, \varepsilon_{Xt}, \varepsilon_{r_{\text{world}t}}\}$$